

Project Design Phase-II

Technology Stack (Architecture & Stack)

Date	04 July 2026
Team ID	
Project Name	Plugging into the Future: An Exploration of Electricity Consumption Patterns
Maximum Marks	4 Marks

Technology Stack — Electricity Consumption Analytics Platform

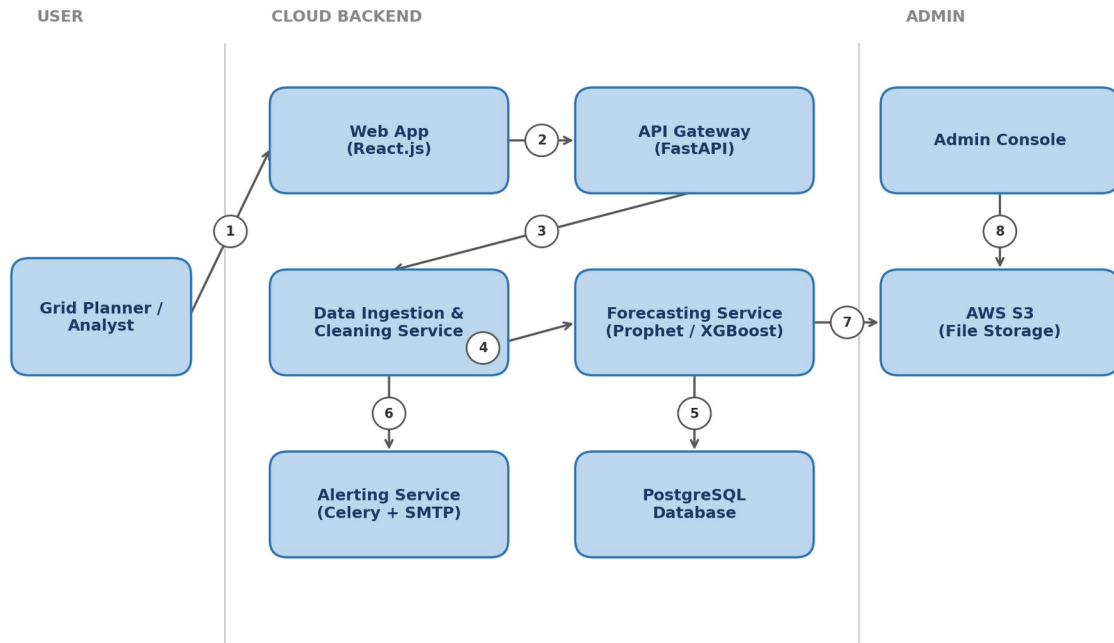


Table-1: Components & Technologies

S.No	Component	Description	Technology
1	User Interface	Dashboard, data upload, and alert views	React.js, Chart.js / Plotly, HTML, CSS
2	Application Logic-1	Data ingestion, cleaning & validation	Python (Pandas), FastAPI
3	Application Logic-2	Demand forecasting engine	Python (Prophet, XGBoost, scikit-learn)
4	Application Logic-3	Alerting & notification service	Python (Celery), SMTP / Twilio
5	Database	Consumption & forecast records	PostgreSQL
6	Cloud Database	Managed relational database service	AWS RDS for PostgreSQL
7	File Storage	Raw CSV uploads & exported reports	AWS S3
8	External API-1	Weather correlation for demand modeling	OpenWeather API
9	External API-2	SMS / email alert delivery	Twilio API / SendGrid API
10	Machine Learning Model	Time-series electricity demand forecasting	Prophet / XGBoost regression model
11	Infrastructure (Server / Cloud)	Application deployment & orchestration	Docker + Kubernetes on AWS EKS

Table-2: Application Characteristics

S.No	Characteristics	Description	Technology
1	Open-Source Frameworks	React, FastAPI, Pandas, scikit-learn, Prophet	Listed frameworks
2	Security	HTTPS/TLS, JWT-based auth, role-based access control, encrypted storage	TLS 1.3, JWT, AES-256
3	Scalable Architecture	Microservices behind a load balancer; each service scales independently	Docker, Kubernetes (AWS EKS)
4	Availability	Multi-AZ deployment with load balancer and auto-scaling groups	AWS Elastic Load Balancer, Auto Scaling
5	Performance	Caching for common dashboard queries; CDN for static assets	Redis cache, CloudFront CDN